A New Method for Fault Location in Distribution Networks Based on Voltage Sag Measurements

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***Abstract*—This paper presents an innovative method for fault location in distribution networks based on classical circuit analyses. Two synchronized and few nonsynchronized pre- and during-fault voltages are required at few buses along with the impedancematrix. A new impedance matrix manipulation procedure is proposed so that the distribution system is investigated in partitions across multiple subsystems. This procedure allows the fault location process to be performed by solving systems of determined equations, making the method more technically accessible. The fault is located by analyzing the voltage sag at the terminal-bus of each subsystem separately. The proposed method is validated on the IEEE 33-bus, 12.66 kV distribution system with/without distributed generation (DG). Simulation results show the robustness and accuracy of the method under several pre- and during-fault scenarios and measurements errors.**

***Index Terms*—Classical technique, fault location, distribution networks, distributed generation, voltage measurements.**

I. Introduction

FAULT location is one of the most important functions in distribution networks. Significantly reducing the restoration time, the fault location process improves the feeder reliability and quality of the electricity supply for the customers [1]. Due to such benefits, several methods for fault location in distribution networks have been proposed based on the information provided by intelligent electronic devices (IEDs) [2]–[21].

Since it is not feasible to install IEDs at all buses of the distribution system, most of the methods have been proposed by adopting measuring devices at a few buses [2]–[20]. The lack of parameters information, caused by the sparse measurements, results in underdetermined systems of equations. To solve this problem, complex mathematical techniques such as state estimation, optimization, artificial neural network, support vector machine, fuzzy logic, and genetic algorithm have been employed [22].

The methods proposed in [3]–[7] use IEDs at a few buses to estimate the fault location in distribution networks. Their performance is not evaluated in a scenario with DGs and load data are required. According to [2], besides the voltage and/or current measuring devices, smart load meters should also be installed for some methods that require load data, such as [3]–[5]. In [6], due to the use of sparse voltage measurements, an iterative load flow calculation is required, which increases the run time. The method proposed in [7] cannot be used for unbalanced three-phase faults, and its performance depends on the accuracy of the load, feeder, and source data.

Although the methods proposed in [8]–[12] also require load data, DG is addressed. However, only inverter-interfaced distributed generations (IIDGs) are considered in [11], [12]. High impedance faults cannot be identified in [11] as well as three-phase fault in [12].

In [13], [14], smart meters (SMs) are placed at few buses of the system to provide nonsynchronized pre-fault and during-fault voltage measurements. From a voltage sag vector, a current sparse vector is calculated. In [13], the higher value of the current sparse vector indicates the faulted bus, while the method proposed in [14] uses k-nearest neighborhood technique to estimate the fault location. Both methods [13], [14] may be more attractive than [3]–[7], since the load data are not required. However, in [13], [14], there is no technique showing guidelines on where measuring devices should be placed, and the DG impacts are not addressed.

In [15], the faulted area of the distribution system is identified based on the voltage measurements provided by SMs. In fact, such a method is used to improve the performance of conventional impedance-based techniques, identifying the area where the voltage magnitude is low due to the proximity of fault location. Although few SMs are required, this method is not evaluated in a scenario with DG [15].

The methods proposed in [16]–[18] identify the faulted section in distribution networks with/without DGs by employing fault indicators (FIs). Such methods do not require load data and can provide accurate location results. However, a considerable number of FIs must be installed. Furthermore, some buses of the distribution system must also be equipped with SMs for the method proposed in [18]. For the IEEE 33-bus test feeder, eight SMs and seven FIs are required [18].

In [19], the substation source and all buses with DG must be equipped with digital fault recorders (DFRs). The fault current value is obtained through the sum of the currents injected by each DG and the substation source. Knowing the impedance matrix of the system and the fault current, different voltage sags are calculated for all buses with DG. Since the actual voltage sag in

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each bus with DG is recorded, the calculated voltage sag with the lowest error, compared to the actual voltage sag, corresponds to the fault location. As described, such a method [19] is based on classical analysis, and does not require complex mathematical techniques and load data.

Similar to [19], the method proposed in [20] assumes that synchronized measurements of pre- and during-fault voltages and currents are obtained from DFRs in the substation and in each DG unit. The pre-fault data are used to estimate the load power demand on each bus of the system through a load-flow, increasing the run time. The impedance matrix and the during-fault voltages are used to identify the fault location. Both [19], [20] are proposed in a scenario with DG, regardless of the DG type and interface. Their performance is not evaluated under measurement error conditions [19], [20].

A PMU based fault location technique for distribution networks using real-time state estimation is presented in [21]. The method estimates the faulted line under noisy conditions, regardless of the DG type, fault type, and fault impedance. For this, all buses of the distribution system must be equipped with PMUs, which makes this method less accessible from a practical point of view.

This paper introduces a new method for fault location in distribution networks with significant contributions, as follows:

- Compared to [3]–[12], the proposed method does not require loads data. Only the voltage sag at a few buses and the impedance matrix must be known.
- Unlike [7] and [11], [12], all fault types are identified in this paper with a wide range of fault impedance.
- The present method is applicable for systems with/without DG, regardless of the DG type. In [3]–[7] and [13]–[15], DGs are not introduced, and only IIDGs are addressed in [11], [12].
- The proposed method requires only two μPMUs and few SMs for whatever the system size, resulting in fewer IEDs in general than [3]–[5], [11], [16]–[18], and [21].
- Unlike [7] and [19], [20], the method proposed in this paper is evaluated by considering errors for the measuring devices and for the impedance matrix.
- DGs are addressed in [2], [19], and [20]. However, a measuring device is required at all buses with DG, which is not required for the method proposed in this paper.
- The present method locates the fault without mathematical complexity. Systems of determined equations are solved by a simple multiplication of matrices and vectors. Thus, besides being a method more accessible to utility professionals, it provides a computational time of less than three seconds.

The content of this paper is organized as follows. In Section II, the proposed method for fault location is described. In Section III, the method performance is evaluated under several pre- and during fault scenarios. Finally, the conclusions are presented in Section IV.

II. New Fault Location Methodology

The proposed method locates the fault based on the formation of several subsystems. During the fault location process, each subsystem is investigated separately. Basically, the subsystems represent the laterals of the distribution system. Thus, the number of subsystems to be created is equal to the number of terminal-buses, where each subsystem consists of the series impedance between the substation-bus and its terminal-bus.

For simplicity, the proposed methodology is described by adopting the 7-bus distribution system shown in Fig. 1(a). All line sections are adopted with the same three-phase impedance Z, and Z_T is the three-phase impedance of the substation transformer. As illustrated in Fig. 1(b), three subsystems are created for the distribution system of Fig. 1(a).

Regardless of the distribution system size, the proposed method requires only two synchronized voltage measurements and S-1 non-synchronized voltage measurements, where S is the number of subsystems. As indicated in Fig. 1, the synchronized and non-synchronized voltage measurements are provided by μPMUs and SMs, respectively. A μPMU is installed at the substation-bus (or other IED that can provide synchronized measurements, such as digital relays) and at any terminal-bus. SMs are installed at the remaining terminal-buses (Fig. 1).

When a fault occurs at section 2-3, for example, as indicated in Fig. 1(a), all subsystems are affected by the fault as shown in Fig. 1(b). Although the faulted section 2-3 is not present in the subsystems 2 and 3, the fault current is injected into such subsystems by the junction-bus-2, i.e., when a subsystem does not contain the faulted section, the fault current is injected at its junction-bus closest to the fault (such affirmation is valid in a scenario with/without DG). Therefore, since the voltage sags caused at the terminal-bus of each subsystem are associated with the same fault current, any subsystem can be adopted to calculate the fault current.

A. Fault Current Calculation

During the fault current calculation, the magnitude and angle of the pre- and during-fault voltages must be known. Hence, the fault current is calculated from the subsystem equipped with μPMU at its terminal-bus. Arbitrarily, a μPMUis placed at the terminal-bus-7. However, as mentioned previously, the fault current could be calculated from any other subsystem with a μPMU at its terminal-bus.

As illustrated in Fig. 2, all buses between the substation-bus and the terminal-bus are eliminated for the subsystem equipped with μPMU at its terminal-bus (subsystem 3). From this, the number of measuring devices is equal to the number of buses, and provides a determined system of equations from which the fault current is calculated. For explanatory purposes, the identification number of the eliminated bus-2 is maintained.

The procedure in Fig. 2 is adopted due to the fault current is significantly higher than the variation of the pre- and during-fault currents of the loads. Furthermore, it must be noted that the series impedance between the substation-bus and the terminal-bus does not change, i.e., the impedance between bus-1 and bus-7 is 2Z for the subsystem 3 in Fig. 1 and Fig. 2.

The voltage sags at buses 1 and 7 are obtained to compose a *(6 x 1)* voltage sag vector. The voltage sag at each bus is given by (1), where $\dot{V}_n^{abc,pf}$ and $\dot{V}_n^{abc,f}$ are, respectively, the three-phase

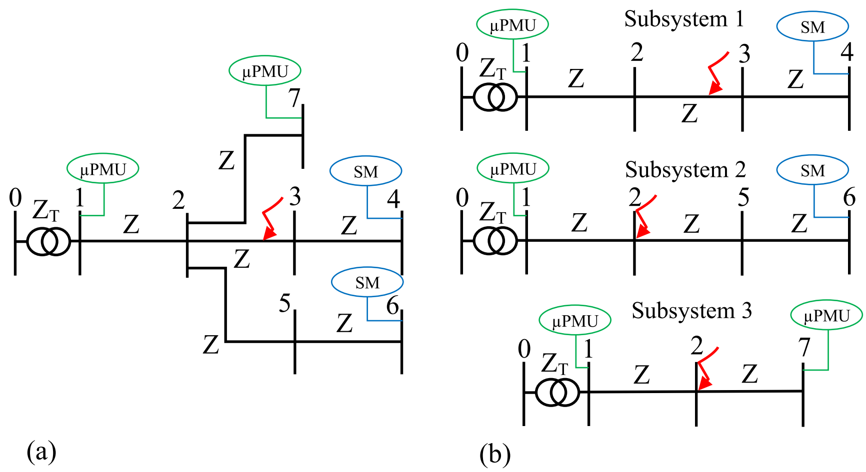

Fig. 1. (a) 7-bus distribution system; (b) Proposed subsystems.

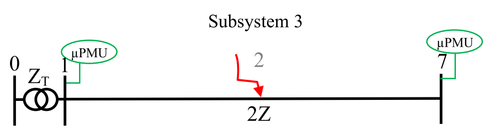

Fig. 2. Subsystem 3 only with bus-1 and bus-7, both equipped with μPMU.

pre- and during-fault voltages at bus-*n*.

$$\Delta\dot{V}_n^{abc} = \begin{bmatrix} \dot{V}_n^{a,pf} - \dot{V}_n^{a,f} \\ \dot{V}_n^{b,pf} - \dot{V}_n^{b,f} \\ \dot{V}_n^{c,pf} - \dot{V}_n^{c,f} \end{bmatrix}_{3x1} \tag{1}$$

The three-phase impedance matrix of the subsystem 3 is built from the topology shown in Fig. 2. By multiplying the inverse of the impedance matrix by the voltage sag vector, a current vector is obtained as described in (2).

$$\begin{bmatrix} \dot{I}_1^{abc} \\ \dot{I}_7^{abc} \end{bmatrix}_{6x1} = inv\begin{bmatrix} Z_T & Z_T \\ Z_T & Z_T + 2Z \end{bmatrix}_{6x6} \cdot \begin{bmatrix} \Delta\dot{V}_1^{abc} \\ \Delta\dot{V}_7^{abc} \end{bmatrix}_{6x1} \tag{2}$$

The fault current ($\dot{I}_f^{abc}$) is given by the sum of both elements in the current vector, as indicated in (3). The fault current ($\dot{I}_f^{abc}$) has one nonzero element, two nonzero elements, and three nonzero elements for single-phase, double-phase, and three-phase faults, respectively.

$$\dot{I}_f^{abc} = \dot{I}_1^{abc} + \dot{I}_7^{abc} \tag{3}$$

It is well known that the DG presence changes the pre- and during-fault voltages when compared to a scenario without DG [23], [24]. Hence, since the fault current is calculated based on the voltage sag, the DG effects are already included in the pre- and during-fault voltages measured by the μPMUs. Thus, the subsystems formation procedure shown in Fig. 2 is the same with/without DG, i.e., all buses between the substation-bus and terminal-bus are eliminated even if a DG is connected. Moreover, the proposed method is insensitive to the fault resistance, as its effect is also included in the measured during-fault voltages.

B. Locating the Bus Closest to the Fault

After calculating the fault current ($\dot{I}_f^{abc}$) by (3), through the subsystem equipped with μPMUs, the fault location process is performed by investigating the magnitude of the voltage sag at the terminal-bus of each subsystem. Hence, measuring devices are required at all terminal-buses, as shown in Fig. 1.

Initially, consider only the subsystem 3. By assuming bus-1 as the faulted bus, a sparse current vector is determined containing the calculated fault current ($\dot{I}_f^{abc}$) at bus-1 position and the other elements set to zero. By multiplying the impedance matrix of the subsystem 3 by the determined sparse current vector, a voltage sag vector is obtained, as (4). The same process is performed for all buses of the subsystem 3, as described in (5) and (6). It must be noted in (4), (5), and (6) that the impedance matrix of the subsystem 3 is built by considering all buses, as illustrated in Fig. 1(b).

From (4) to (6), three different voltage sags are calculated for the terminal-bus-7 ($\Delta\dot{V}_{7,n}^{abc,i}$), where $\Delta\dot{V}_{m,n}^{abc,i}$ is the three-phase voltage sag calculated at terminal-bus-*m* due to a fault current injection at bus-*n*. *i* represents the subsystem under analysis.

$$\begin{bmatrix} \Delta\dot{V}_{1,1}^{abc,3} \\ \Delta\dot{V}_{2,1}^{abc,3} \\ \Delta\dot{V}_{7,1}^{abc,3} \end{bmatrix}_{9x1} = \begin{bmatrix} Z_T & Z_T & Z_T \\ Z_T & Z_T + Z & Z_T + Z \\ Z_T & Z_T + Z & Z_T + 2Z \end{bmatrix}_{9x9} \cdot \begin{bmatrix} \dot{I}_f^{abc} \\ 0 \\ 0 \end{bmatrix}_{9x1} \tag{4}$$

$$\begin{bmatrix} \Delta\dot{V}_{1,2}^{abc,3} \\ \Delta\dot{V}_{2,2}^{abc,3} \\ \Delta\dot{V}_{7,2}^{abc,3} \end{bmatrix}_{9x1} = \begin{bmatrix} Z_T & Z_T & Z_T \\ Z_T & Z_T + Z & Z_T + Z \\ Z_T & Z_T + Z & Z_T + 2Z \end{bmatrix}_{9x9} \cdot \begin{bmatrix} 0 \\ \dot{I}_f^{abc} \\ 0 \end{bmatrix}_{9x1} \tag{5}$$

$$\begin{bmatrix} \Delta\dot{V}_{1,7}^{abc,3} \\ \Delta\dot{V}_{2,7}^{abc,3} \\ \Delta\dot{V}_{7,7}^{abc,3} \end{bmatrix}_{9x1} = \begin{bmatrix} Z_T & Z_T & Z_T \\ Z_T & Z_T + Z & Z_T + Z \\ Z_T & Z_T + Z & Z_T + 2Z \end{bmatrix}_{9x9} \cdot \begin{bmatrix} 0 \\ 0 \\ \dot{I}_f^{abc} \end{bmatrix}_{9x1} \tag{6}$$

Assuming the module of the voltage sag measured at the terminal-bus-7 ($\Delta\dot{V}_7^{abc}$) as reference, an error ($Error_{7,n}^{abc,i}$) is attributed for each voltage sag calculated at the terminal-bus-7 ($\Delta\dot{V}_{7,n}^{abc,i}$), as described in (7).

This process, from (4) to (7), is performed for all subsystems. Hence, the error vector results as (8) and (9) for the subsystems 1 and 2, respectively, where $Error_{m,n}^{abc,i}$ is the error attributed for the voltage sag calculated at the terminal-bus-*m* of the subsystem *i* when the fault current is injected at bus-*n*.

$$Error_7^{abc,3} = \begin{bmatrix} Error_{7,1}^{abc,3} \\ Error_{7,2}^{abc,3} \\ Error_{7,7}^{abc,3} \end{bmatrix} = \begin{bmatrix} \left|\Delta\dot{V}_7^{abc}\right| \\ \left|\Delta\dot{V}_7^{abc}\right| \\ \left|\Delta\dot{V}_7^{abc}\right| \end{bmatrix} - \begin{bmatrix} \left|\Delta\dot{V}_{7,1}^{abc,3}\right| \\ \left|\Delta\dot{V}_{7,2}^{abc,3}\right| \\ \left|\Delta\dot{V}_{7,7}^{abc,3}\right| \end{bmatrix} \tag{7}$$

$$Error_4^{abc,1} = \begin{bmatrix} Error_{4,1}^{abc,1} \\ Error_{4,2}^{abc,1} \\ Error_{4,3}^{abc,1} \\ Error_{4,4}^{abc,1} \end{bmatrix} = \begin{bmatrix} \left|\Delta\dot{V}_4^{abc}\right| \\ \left|\Delta\dot{V}_4^{abc}\right| \\ \left|\Delta\dot{V}_4^{abc}\right| \\ \left|\Delta\dot{V}_4^{abc}\right| \end{bmatrix} - \begin{bmatrix} \left|\Delta\dot{V}_{4,1}^{abc,1}\right| \\ \left|\Delta\dot{V}_{4,2}^{abc,1}\right| \\ \left|\Delta\dot{V}_{4,3}^{abc,1}\right| \\ \left|\Delta\dot{V}_{4,4}^{abc,1}\right| \end{bmatrix} \tag{8}$$

$$Error_6^{abc,2} = \begin{bmatrix} Error_{6,1}^{abc,2} \\ Error_{6,2}^{abc,2} \\ Error_{6,5}^{abc,2} \\ Error_{6,6}^{abc,2} \end{bmatrix} = \begin{bmatrix} \left|\Delta\dot{V}_6^{abc}\right| \\ \left|\Delta\dot{V}_6^{abc}\right| \\ \left|\Delta\dot{V}_6^{abc}\right| \\ \left|\Delta\dot{V}_6^{abc}\right| \end{bmatrix} - \begin{bmatrix} \left|\Delta\dot{V}_{6,1}^{abc,2}\right| \\ \left|\Delta\dot{V}_{6,2}^{abc,2}\right| \\ \left|\Delta\dot{V}_{6,5}^{abc,2}\right| \\ \left|\Delta\dot{V}_{6,6}^{abc,2}\right| \end{bmatrix} \tag{9}$$

For each subsystem, different errors are obtained when the fault current is injected at all buses individually. The bus associated with the lowest error of each subsystem is selected and called as candidate bus. The candidate bus of each subsystem is used to compose the vector *X* described in (10). As *S* subsystems are created, *S* buses are selected as candidate buses, where x_i is the bus selected as candidate bus in the subsystem *i*.

$$X = \begin{bmatrix} x_1 \; x_2 \cdots x_i \cdots x_{S-1} \; x_S \end{bmatrix} \tag{10}$$

Considering the fault occurrence at section 2-3, as shown in Fig. 1(a), the lowest error of each subsystem will result for its bus closer to the fault. For the analyses, it is considered that the fault occurs closest to bus-3. Hence, bus-3 is selected as candidate bus of the subsystem 1.

As can be seen in Fig. 1(a), the faulted section 2-3 is not present in the subsystems 2 and 3. As mentioned previously, when a subsystem does not contain the faulted section, the fault current is injected at its junction-bus closest to the fault. Hence, the lowest error will be associated with such a junction-bus. In this case, bus-2 is selected as the candidate bus in both subsystems 2 and 3. Therefore, when a fault occurs at section 2-3, closest to bus-3, the vector *X* results as (11).

$$X = \begin{bmatrix} 3 & 2 & 2 \end{bmatrix} \tag{11}$$

TABLE I
CANDIDATE BUS OF EACH SUBSYSTEM FOR DIFFERENT FAULT DISTANCES

Faulted Section	Fault Distance (%)	Candidate bus of each subsystem x_1	x_2	x_3
1-2	33	1	1	1
	67	2	2	2
2-3	33	2	2	2
	67	3	2	2
3-4	33	3	2	2
	67	4	2	2
2-5	33	2	2	2
	67	2	5	2
5-6	33	2	5	2
	67	2	6	2
2-7	33	2	2	2
	67	2	2	7

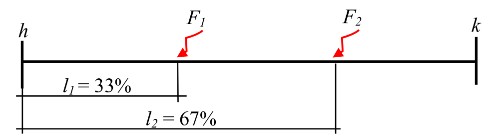

Fig. 3. Section *h*-*k* under two different fault locations.

For a special case, when a fault occurs exactly in the middle of section 2-3, the lowest error of subsystem 1 will be for buses 2 and 3. Under such a condition, any of the buses can be selected as candidate bus, as both are connected at the faulted section 2-3.

In Table I, the candidate bus of each subsystem is shown when a fault occurs at all sections of the 7-bus distribution system of Fig. 1(a). For each section, two different fault locations are considered, as illustrated in Fig. 3.

In Fig. 3, a section between buses *h* and *k* is illustrated, where l_1 and l_2 represent the distances between the upstream bus (bus-*h*) and the fault locations F_1 and F_2, respectively. Such fault distances are given by the section length percentage.

The vector *X* indicates the candidate bus of each subsystem. However, it is necessary to identify which candidate bus, in (10), is closest to the fault. For this, three additional vectors are created, such as vectors *Y*, *J*, and *W*.

The element y_i of the vector *Y*, in (12), indicates how many subsystems the candidate bus-x_i of the vector *X*, in (10), is present. Bus-3, in (11), is present only in subsystem 1 (one subsystem), and bus-2 in all subsystems 1, 2, and 3 (three subsystems). Therefore, from (11), *Y* results as (13).

$$Y = \begin{bmatrix} y_1 \; y_2 \cdots y_i \cdots y_{S-1} \; y_S \end{bmatrix} \tag{12}$$

$$Y = \begin{bmatrix} 1 & 3 & 3 \end{bmatrix} \tag{13}$$

The element j_i of the vector *J*, in (14), indicates how many times bus-x_i, in (10), has been selected as candidate bus. Bus-3, in (11), has been selected as candidate bus only in subsystem 1 (one time), and bus-2 has been selected in subsystems 2 and 3 (two times). Therefore, from (11), the vector *J* results as (15).

$$J = \begin{bmatrix} j_1 \; j_2 \cdots j_i \cdots j_{S-1} \; j_S \end{bmatrix} \tag{14}$$

$$J = \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} \tag{15}$$

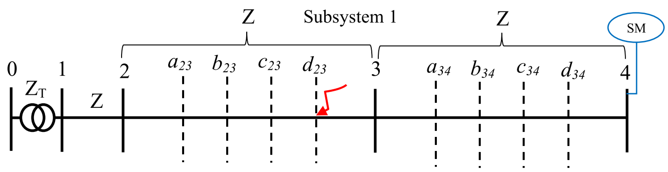

Fig. 4. Subsystem 1 with fictitious buses.

Through the vectors Y and J, the weight vector (W) is calculated by (16). Ideally, the bus closest to the fault should be selected as candidate bus in all subsystems in which it is present. Therefore, the highest weight in (16) indicates which candidate bus in (10) is closest to the fault, where $0 < w_i \leq 1$. For the vector X obtained in (11), the vector W results as (17), indicating bus-3 as the bus closest to the fault.

$$W = \begin{bmatrix} \frac{j_1}{y_1} & \frac{j_2}{y_2} & \cdots & \frac{j_i}{y_i} & \cdots & \frac{j_{S-1}}{y_{S-1}} & \frac{j_S}{y_S} \end{bmatrix} \tag{16}$$

$$W = \begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \end{bmatrix} \tag{17}$$

C. Fault Location

After identifying the bus closest to the fault, the faulted section can be located, as well as the faulted point. For this, fictitious buses (dashed buses in Fig. 4) are added to all sections connected to bus-3 (bus selected as the bus closest to the fault). Since bus-3 is present only in subsystem 1, such a procedure is applied only for subsystem 1.

The impedance matrix of subsystem 1 must be built by considering the topology presented in Fig. 4. From this, the process described from (4) to (9) must be repeated only for the subsystem 1. The fictitious bus presenting the lowest error will indicate the faulted section and the faulted point. Furthermore, the position of the fictitious bus indicates the fault distance.

D. Fault Location for Special Cases

Junction-buses, such as bus-2 in the distribution system in Fig. 1, will be present in more than one subsystem. Hence, when a fault occurs close to a junction-bus, it will be present in more than one position in the vector X. If the junction-bus is selected as the bus closest to the fault (the highest value in the vector W), fictitious buses must be added to all subsystems that contain the junction-bus.

For example, when a fault occurs closest to bus-2 in section 2-3, bus-2 is selected as the bus closest to the fault. In this case, all subsystems must be considered with fictitious buses, as illustrated in Fig. 5. Furthermore, as bus-2 was indicated as the bus closest to the fault, the fault may have occurred exactly at bus-2 or at any section connected to bus-2. Therefore, fictitious buses must be added to all sections connected to bus-2.

By repeating the process from (4) to (9) for all subsystems with the topologies in Fig. 5 (with fictitious buses), a new vector X is obtained, as (18). As the fault occurred closest to bus-2 in section 2-3, the fictitious bus-a_{23} is selected as candidate bus in subsystem 1, and bus-2 in subsystems 2 and 3. The new vectors Y, J, and W also result as (13), (15), and (17). Thus, bus-a_{23} is

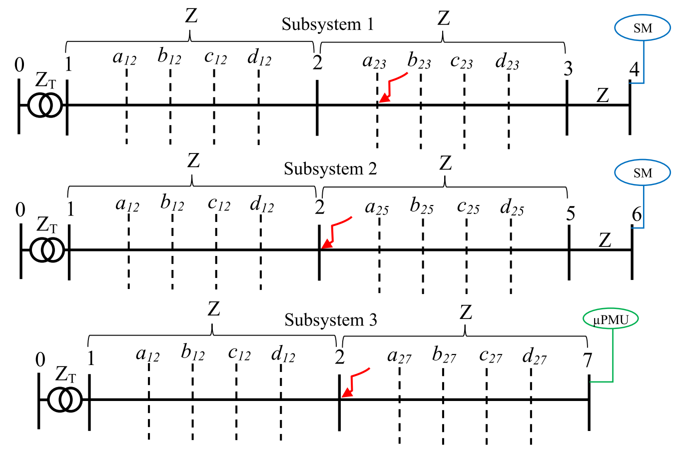

Fig. 5. Subsystems 1, 2, and 3 with fictitious buses at different sections.

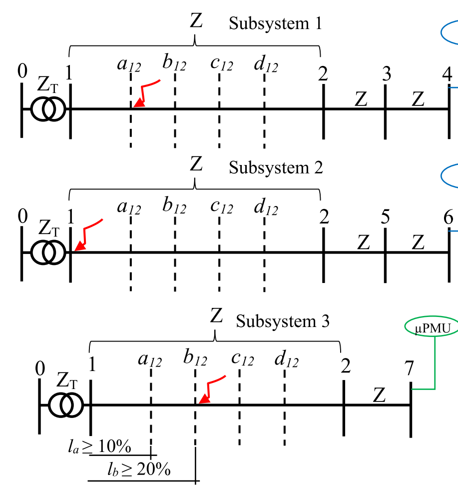

Fig. 6. Subsystems 1, 2, and 3 with fictitious buses in the same section 1-2.

identified as the fault location.

$$X = \begin{bmatrix} a_{23} & 2 & 2 \end{bmatrix} \tag{18}$$

Another special case refers to when a fault occurs in a section present in more than one subsystem. For example, when a fault occurs at bus-a_{12} in section 1-2 (Fig. 6). Ideally, the faulted bus-a_{12} should be selected as candidate bus in all subsystems in which it is present. However, as illustrated in Fig. 6, adjacent buses such as bus-1 and bus-b_{12} may also be selected as candidate buses. Thus, the new vector X would be composed by bus-1, bus-a_{12}, and bus-b_{12}. Hence, three different fault locations would be identified by the new vector W.

Such a condition may occur due to the proximity between adjacent buses. In this case, the pre-fault voltage at the terminal-bus of each subsystem is investigated. The bus selected in the subsystem with the highest voltage profile is assumed as the faulted point.

To decrease the possibility of selecting different buses in the same section, the fictitious buses should be added so that a distance of at least 10% is obtained between all adjacent buses,

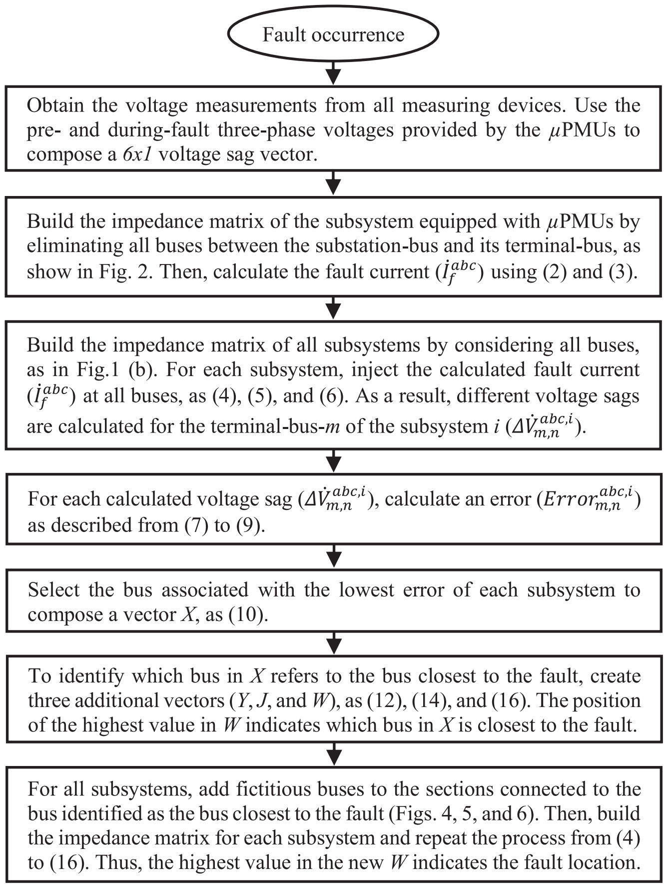

Fig. 7. Flowchart of the proposed method for fault location.

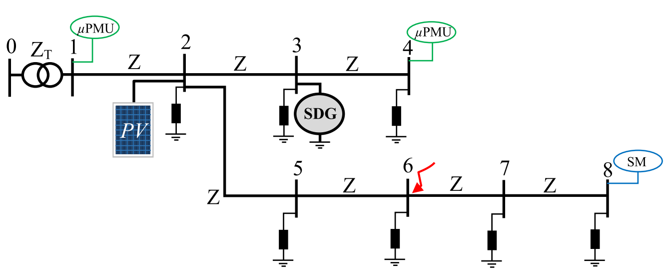

Fig. 8. Diagram of the 8-bus, 13.8 kV distribution system.

as illustrated in subsystem 3 of Fig. 6. The proposed method is described in detail in Fig. 7.

E. Numerical Calculation Procedure

In order to present a numerical calculation procedure, the proposed method is applied to the 8-bus distribution system of Fig. 8. A 500 kVA, 69/13.8 kV three-phase power transformer is placed at the substation with a resistance of 1% and a reactance of 3%. All line sections are adopted with a length of 100 meters and the same three-phase impedance Z, where the diagonal and off-diagonal elements are 0.8265 + j0.8364 Ω/km and 0.1305 + j0.3187 Ω/km, respectively. A synchronous DG of 300 kVA is placed at bus-3 and a 200 kW PV at bus-2. At all buses, a load

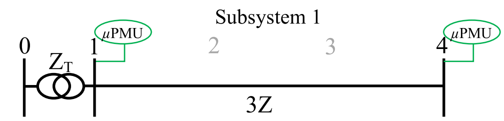

Fig. 9. Subsystem configuration used during the fault current calculation.

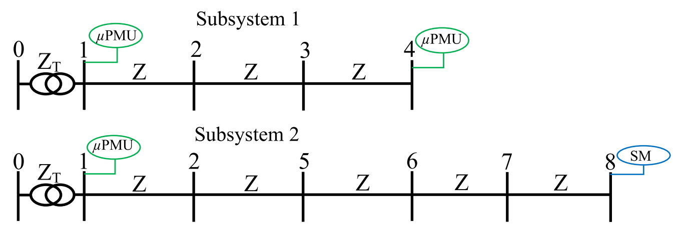

Fig. 10. Subsystem configuration used during the fault current injection.

of 70 kW is placed as constant-power model. The distribution system is modeled in DIgSILENT software [25].

When a LLL fault occurs at bus-6 with a fault resistance of 10 ohms, the synchronized voltage sag phasors at buses 1 and 4 are used to calculate the fault current. For this, the subsystem configuration of Fig. 9 is used.

Therefore, from the subsystem of Fig. 9, the fault current is calculated by (19). For simplicity, only phase A is illustrated in the calculation procedure.

$$\begin{bmatrix} 233.6 - \mathrm{j}180.2 \\ 102.7 - \mathrm{j}74.7 \end{bmatrix}_{6x1} = inv \begin{bmatrix} Z_T & Z_T \\ Z_T & Z_T + 3Z \end{bmatrix}_{6x6} \cdot \begin{bmatrix} 4182.0 + \mathrm{j}2865.2 \\ 4215.1 + \mathrm{j}2865.5 \end{bmatrix}_{6x1} \quad (19)$$

From (19), the fault current is given by (20).

$$\dot{I}_f^a = (233.6 - \mathrm{j}180.2) + (102.7 - \mathrm{j}74.7) = 336.4 - \mathrm{j}254.9\mathrm{A} \quad (20)$$

Then, the calculated fault current is injected at all buses of each subsystem as described from (4) to (6). For this, the configurations of Fig. 10 are used.

After injecting the fault current at each bus of both subsystems, different voltage sags are calculated at buses 4 and 8. Then, an error is attributed for all calculated voltage sags as described from (7) to (9). For this, the magnitude of the actual voltage sag at buses 4 and 8 must be obtained. Therefore, the voltage sag error of each bus of subsystems 1 and 2 is given in (21) and (22), respectively.

$$\begin{bmatrix} Error_{4,1}^{abc,1} \\ Error_{4,2}^{abc,1} \\ Error_{4,3}^{abc,1} \\ Error_{4,4}^{abc,1} \end{bmatrix} = \begin{bmatrix} 33.05 \\ 3.62 \\ 40.18 \\ 76.79 \end{bmatrix} \quad (21)$$

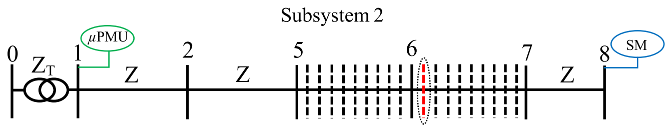

Fig. 11. Subsystem 2 with fictitious buses.

$$\begin{bmatrix} Error_{8,1}^{abc,2} \\ Error_{8,2}^{abc,2} \\ Error_{8,5}^{abc,2} \\ Error_{8,6}^{abc,2} \\ Error_{8,7}^{abc,2} \\ Error_{8,8}^{abc,2} \end{bmatrix} = \begin{bmatrix} 116.67 \\ 80.07 \\ 43.49 \\ 7.29 \\ 29.92 \\ 66.47 \end{bmatrix} \quad (22)$$

From (21), it is noted that the lowest error of subsystem 1 is obtained for bus-2. From (22), the lowest error of subsystem 2 is obtained for bus-6. Hence, buses 2 and 6 are selected as a candidate bus of subsystems 1 and 2, respectively. Therefore, the vector *X* results as (23).

$$X = \begin{bmatrix} 2 & 6 \end{bmatrix} \quad (23)$$

The candidate bus-2 is present in subsystem 1 and subsystem 2 (two subsystems). The candidate bus-6 is present only in subsystem 2 (one subsystem). Thus, the vector *Y* results as (24).

$$Y = \begin{bmatrix} 2 & 1 \end{bmatrix} \quad (24)$$

The candidate bus-2 was selected as a candidate bus only in subsystem 1 (one time). The candidate bus-6 was selected as a candidate bus only in subsystem 2 (one time). Therefore, the vector *J* results as (25).

$$J = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad (25)$$

Finally, from (24) and (25), the vector *W* results as (26).

$$W = \frac{J}{Y} = \begin{bmatrix} \frac{1}{2} & \frac{1}{1} \end{bmatrix} \quad (26)$$

The highest value in vector (26) indicates bus-6 as the bus closest to the fault. Since bus-6 is present only in subsystem 2, fictitious buses are added only to subsystem 2 as shown in Fig. 11. Since bus-6 was selected as the bus closest to the fault, fictitious buses must be added to all sections connected to bus-6. As illustrated in Fig. 11, 9 fictitious buses were added, resulting in a distance of 10% between all buses.

The same procedure performed in (22) must be repeated for subsystem 2 with the topology of Fig. 11 (with fictitious buses). In this case, an error will be calculated for all buses between bus-5 and bus-7. The lowest error will indicate the fault location. For the case under analysis, the first fictitious bus of section 6-7 presented the lowest error. Hence, the method estimated the fault location at 10% of section 6-7. Therefore, since section 6-7 has a distance of 100 meters, an error of only 10 meters was obtained.

III. Method Validation

DIgSILENT software is utilized to model the distribution network. The performance of the proposed method is validated on the IEEE 33-bus, 12.66 kV distribution network presented in Fig. 12. The system data are available in [26]. A 10 MVA three-phase transformer is used in the substation with a resistance and reactance of 1% and 7%, respectively. The loads were modeled as constant-power loads.

TABLE II
Fault Simulation Results Under Different Fault Resistances

Faulted Section	Fault Type	Fault Resistance (Ω)	Fault Distance (%)	Indicated Faulted Section	Indicated Distance (%)
1-2	LG	1 – 100	33 / 67	1-2 / 1-2	22 / 56
2-3	LG	1 – 100	33 / 67	2-3 / 2-3	22 / 67
3-4	LG	1 – 100	33 / 67	3-4 / 3-4	22 / 56
4-5	LG	1 – 100	33 / 67	4-5 / 4-5	11 / 56
5-6	LG	1 – 100	33 / 67	5-6 / 5-6	22 / 56
6-7	LG	1 – 100	33 / 67	6-7 / 6-7	22/ 56
7-8	LG	1 – 100	33 / 67	7-8 / 7-8	22 / 56
8-9	LG	1 – 100	33 / 67	8-9 / 8-9	33 / 67
9-10	LG	1 – 100	33 / 67	9-10 / 9-10	33 / 67
10-11	LG	1 – 100	33 / 67	10-11 / 10-11	11 / 44
11-12	LG	1 – 100	33 / 67	11-12 / 11-12	22 / 56
12-13	LG	1 – 100	33 / 67	12-13 / 12-13	33 / 67
13-14	LG	1 – 100	33 / 67	13-14 / 13-14	33 / 67
14-15	LG	1 – 100	33 / 67	14-15 / 14-15	33 / 67
15-16	LG	1 – 100	33 / 67	15-16 / 15-16	33 / 78
16-17	LG	1 – 100	33 / 67	16-17 / 16-17	33 / 67
17-18	LG	1 – 100	33 / 67	17-18 / 17-18	44 / 78
2-19	LG	1 – 100	33 / 67	2-19 / 2-19	33 / 67
19-20	LG	1 – 100	33 / 67	19-20 / 19-20	33 / 67
20-21	LG	1 – 100	33 / 67	20-21 / 20-21	33 / 67
21-22	LG	1 – 100	33 / 67	21-22 / 21-22	33 / 67
3-23	LG	1 – 100	33 / 67	3-23 / 3-23	33 / 67
23-24	LG	1 – 100	33 / 67	23-24 / 23-24	33 / 67
24-25	LG	1 – 100	33 / 67	24-25 / 24-25	33 / 67
6-26	LG	1 – 100	33 / 67	Bus-6 / 6-26	0 / 33
26-27	LG	1 – 100	33 / 67	26-27 / 26-27	11 / 44
27-28	LG	1 – 100	33 / 67	27-28 / 27-28	33 / 67
28-29	LG	1 – 100	33 / 67	28-29 / 28-29	22 / 67
29-30	LG	1 – 100	33 / 67	29-30 / 29-30	22 / 56
30-31	LG	1 – 100	33 / 67	30-31 / 30-31	33 / 67
31-32	LG	1 – 100	33 / 67	31-32 / 31-32	33 / 67
32-33	LG	1 – 100	33 / 67	32-33 / 32-33	33 / 78

Two μPMUs and three SMs are required, resulting in 5 measuring devices. A μPMU is placed arbitrarily at bus-18. The method is validated under several pre- and during fault scenarios. For each scenario, two faults were simulated at all sections (at 33% and 67% of each section) as illustrated in Fig. 3, resulting in 64 fault cases. During the faulted point location, 8 fictitious buses were added to the sections under analysis, resulting in a distance of 11.11% between all adjacent buses.

A. Method Sensitivity to the Fault Resistance

In Table II, single-line to ground fault (LG) is addressed. For all cases, the same results were obtained when the fault resistance ranges from 1 to 100 ohms (it is was used 1, 50, and 100 Ω), showing that the method is insensitive to the fault resistance.

As can be seen in Table II, the faulted section was indicated correctly in 63 cases (98.44% of the cases). Only in a case, when a fault occurs at 33% of section 6-26, bus-6 was indicated incorrectly as the faulted point. In this case, since bus-6 is connected to three different sections, it is not possible to identify the actual faulted section. However, all three sections refer to

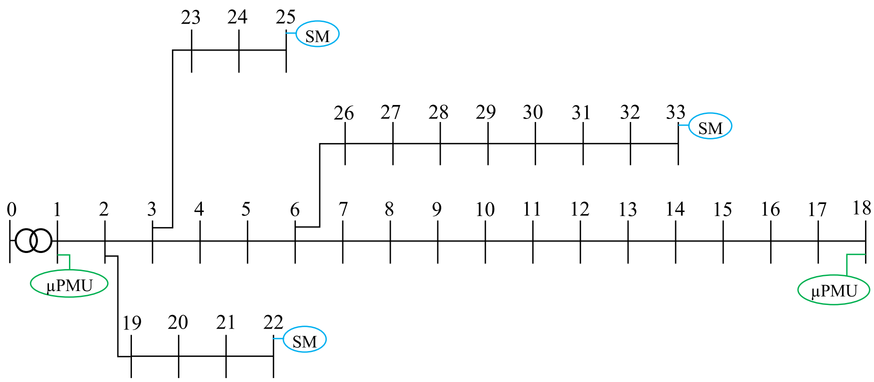

Fig. 12. Diagram of the 33-bus, 12.66 kV distribution system.

TABLE III
METHOD SENSITIVITY TO THE FAULT RESISTANCE

Fault type	Fault resistance	Correct section	Adjacent Section	Correct half	Exact point
LG	1 Ω	63	1	60	36
LG	50 Ω	63	1	60	36
LG	100 Ω	63	1	60	36

TABLE IV
SIMULATION RESULTS UNDER UNBALANCED SYSTEM CONDITION

Fault type	Fault resistance	Correct section	Adjacent Section	Correct half	Exact point
LLL	50 Ω	62	2	57	30
LL	50 Ω	62	2	55	33
LLG	50 Ω	62	2	58	30
AG	50 Ω	63	1	60	35
BG	50 Ω	63	1	59	32
CG	50 Ω	63	1	60	36

TABLE V
SIMULATION RESULTS UNDER LOADING ALTERATION

Total Load	Fault type	Fault resistance	Correct section	Adjacent Section	Correct half	Exact point
70%	LG	5 Ω	62	2	59	41
100%	LG	5 Ω	63	1	60	36
130%	LG	5 Ω	61	3	57	28

TABLE VI
ERRORS ATTRIBUTED ARBITRARILY TO SOME SECTIONS IMPEDANCE

Section	Impedance change percentage
1-2	+5%
7-8	-5%
10-11	+5%
11-12	-5%
17-18	+5%
20-21	+5%
24-25	-5%
27-28	+5%
30-31	-5%

adjacent sections. Thus, it is assumed that the algorithm indicates an adjacent section of the actual faulted section.

Besides indicating the faulted section accurately, the proposed method indicates in which half of the section the fault is located. When the indicated distance is $\leq$44%, it means that the fault occurred in the first half (between 1% and 50% of the section length). When the indicated distance is $\geq$56%, it means that the fault occurred in the second half (between 50% and 100% of the section length). Therefore, from 63 sections indicated correctly, 60 were indicated in the correct half (95.24%). Finally, from 60 half sections indicated accurately, 36 refer to the exact point of the fault (60%). From these analyses, Table II can be summarized in Table III.

B. *Method Sensitivity Under Unbalanced System Condition*

In order to investigate the method sensitivity under an unbalanced system condition, the loads present in phase A and phase C were decreased and increased by 20%, respectively. The loads present in phase B were maintained in 100%. The results, in Table IV, show the robustness of the method under an unbalanced system condition for all fault types.

C. *Method Sensitivity Under Different Loading Conditions*

In Table V, the simulation results are shown when all loads are increased and decreased by 30%. The method sensitivity can be verified by analyzing the exact point of the fault. Even under high loading conditions, the correct section was indicated accurately. In the worst case, 95.31% of the sections were indicated correctly, with 93.44% representing the correct half.

D. *Method Sensitivity by considering Line Parameter Errors*

Line parameters in distribution networks may be obtained with errors. From this, the method sensitivity is investigated by attributing errors of $+-5\%$ to the impedance of some sections, as indicated in Table VI.

Assuming the errors in Table VI, the simulation results are described in Table VII. By comparing Tables III and VII, for LG fault type, it is verified that when the impedance matrix of each subsystem is constructed based on incorrect data, the exact point location is affected. However, the correct section and correct half are indicated accurately with 96.87% and 87%, respectively.

TABLE VII
SIMULATION RESULTS UNDER LINE PARAMETER ERRORS

Fault type	Fault resistance	Correct section	Adjacent Section	Correct half	Exact point
LLL	10 Ω	62	2	54	16
LL	10 Ω	62	2	54	16
LLG	10 Ω	62	2	56	22
LG	10 Ω	63	1	58	24

TABLE VIII
ERROR OF THE VOLTAGE MEASURING DEVICES

Measurement devices	Voltage errors
At bus-1 (μPMU)	+0.5%
At bus-18 (μPMU)	-0.5%
At bus-22 (SM)	+0.5%
At bus-25 (SM)	-0.5%
At bus-33 (SM)	+0.5%

TABLE IX
SIMULATION RESULTS UNDER VOLTAGE MEASUREMENT ERRORS CONDITION

Fault type	Fault resistance	Correct section	Adjacent Section	Correct half	Exact point
LLL	100 Ω	59	5	52	19
LL	100 Ω	59	5	50	25
LLG	100 Ω	59	5	52	17
LG	100 Ω	59	5	52	15

TABLE X
SIMULATION RESULTS UNDER VOLTAGE MEASUREMENT ERRORS CONDITION

Fault type	Fault resistance	Correct section	Adjacent Section	Correct half	Exact point
LLL	100 Ω	62	2	57	20

E. Method Sensitivity Under Voltage Measurement Errors

According to [2], [12], and [15], the error of voltage measuring devices ranges from 0.1–0.5%. Thus, the worst error condition is assumed for all IEDs as presented in Table VIII.

In Table IX, simulation results are shown by considering the errors described in Table VIII. Although the voltage measurement errors also affect the exact point location, 92.19% of the sections were indicated correctly. From 59 correct sections, 50 (worst case) were indicated in the correct half, representing 84.75%.

It is important to note that the results in Table IX were obtained when the μPMUs at bus-1 and bus-18 have an error of $+0.5\%$ and -0.5%, respectively. In this case, the fault current is calculated under an error of 1%. In Table X, the simulation results are presented, only for LLL fault, when an error of $+0.5\%$ is also adopted for the μPMU placed at bus-18. In this condition, the method was less affected.

F. Method Sensitivity Under DG Presence

As illustrated in Fig. 13, six DGs are added arbitrarily to the IEEE 33-bus, 12.66 kV distribution system. The capacity of each DG, location, and type are indicated in Table XI.

The method performance with DGs is specified in Table XII. Even under high DG penetration, the method is accurate. In the worst case, 90.63% of the sections were indicated correctly. From 58 correct sections, 45 were indicated in the correct half.

LG fault is the most common fault in distribution networks [22]. As can be seen in Table XII, the method is further robust for this fault type. The correct section was indicated accurately in 98.44% of the fault cases. From 63 correct sections, 85.71% were indicated in the correct half.

TABLE XI
DISTRIBUTED GENERATORS INSTALLED IN THE 33-BUS SYSTEM

Bus	DG type	Capacity
5	Photovoltaic (PV)	250 kW
12	Photovoltaic (PV)	550 kW
18	Asynchronous (ADG)	100 kVA
19	Asynchronous (ADG)	300 kVA
24	Synchronous (SDG)	500 kVA
32	Synchronous (SDG)	100 kVA

TABLE XII
SIMULATION RESULTS UNDER DG PRESENCE

Fault type	Fault resistance	Correct section	Adjacent Section	Correct half	Exact point
LLL	1 Ω	59	5	45	15
LL	1 Ω	58	6	45	15
LLG	1 Ω	62	2	53	11
LG	1 Ω	63	1	54	22

TABLE XIII
DISTRIBUTED GENERATORS INSTALLED IN THE 69-BUS SYSTEM

Bus	DG type	Capacity
16	Photovoltaic (PV)	300 kW
55	Photovoltaic (PV)	250 kW
67	Photovoltaic (PV)	150 kW
35	Asynchronous (ADG)	50 kVA
40	Asynchronous (ADG)	100 kVA
45	Asynchronous (ADG)	150 kVA
27	Synchronous (SDG)	200 kVA
64	Synchronous (SDG)	300 kVA

TABLE XIV
SIMULATION RESULTS FOR THE 69-BUS SYSTEM

DG	Fault type	Fault resistance	Correct section	Adjacent Section	Correct half	Exact point
No	LG	1-100 Ω	126	10	119	78
	LLL		125	11	113	64
Yes	LG	1-100 Ω	120	16	106	67
	LLL		108	28	90	49

G. Method Application for the 69-Bus System

In order to further investigate the robustness of the proposed method, simulations are also performed on the IEEE 69-bus, 12.66 kV distribution system shown in Fig. 14 [27]. Arbitrarily, eight DGs, described in Table XIII, are installed. Two faults are also considered for each section of the 69-bus system (Fig. 3), resulting in 136 cases for each scenario.

In Table XIV, for the 69-bus system, it can be verified that the method is more robust without DG. However, very good location results are also obtained under DG presence. The percentage of correct sections ranges from 80% to 92.65%. From Tables XII

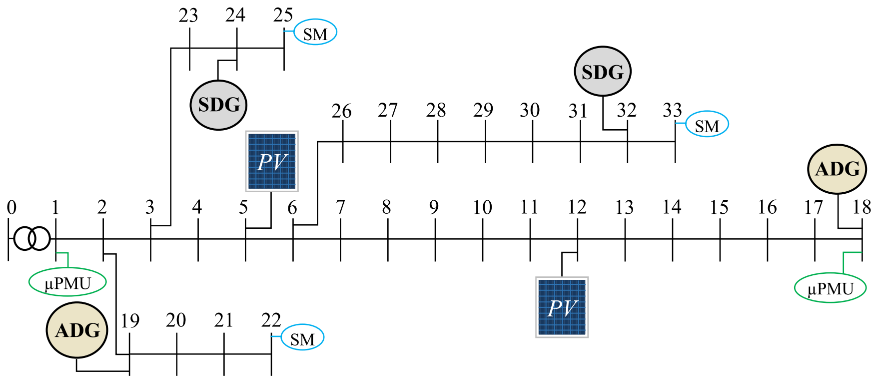

Fig. 13. Diagram of the 33-bus, 12.66 kV distribution system with DGs.

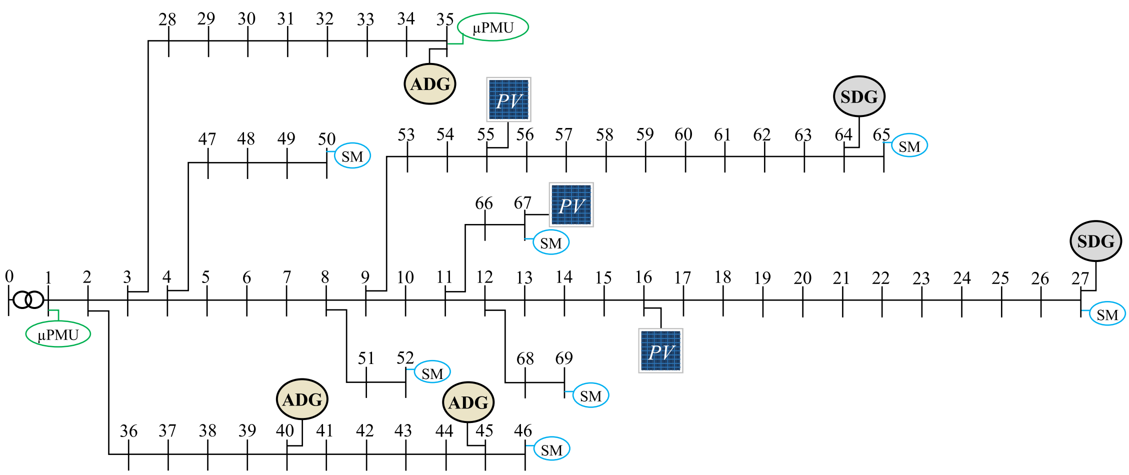

Fig. 14. Diagram of the 69-bus, 12.66 kV distribution system.

and XIV, similar results are verified for different systems. Only correct and adjacent sections are identified.

The performance of the method proposed in [12] is also evaluated on the 69-bus system, where many fault simulation results are presented when faults are also simulated at 33% and 67% of all line sections. When compared to the method in [12], the method introduced in this paper presents very similar results even under DG presence (Table XIV). Moreover, besides indicating the faulted section, this paper also presents very good results when indicating the correct half of the faulted section.

H. Method Application for the 134-Bus System

In order to investigate the performance of the method for a larger system, simulations were performed for the 134-bus, 13.8 kV distribution network illustrated in Fig. 15. The system data are available in [4].

The 134-bus system has many terminal-buses with a short distance from a junction-bus, such as buses 3, 20, 31, 34, 39, 41, 45, 58, 60, 70, 87, 88, 93, 94, 96, 101, 114, 124, and 131. Therefore, in order to reduce the number of measuring devices allocation, subsystems were not created with such terminal-buses. For example, a measuring device was not placed at bus-60, since the distance between the terminal-bus-60 and the nearest junction-bus-52 is very short (the nearest junction-bus to be considered must be present in at least one subsystem; junction-bus-57, for example, is not considered since it is not present in a subsystem). In this case, when a fault occurs anywhere between buses 52 and 60, bus-52 will be indicated as the fault location. Bus-52 will always be indicated as the fault location because it is present in the subsystem formed by bus-55, i.e., when a fault occurs anywhere between buses 52 and 60, bus-52 will refer to the bus closest to the fault. Since the distance between bus-52 and bus-60 is 180 meters, the maximum error for this region is 180 meters. As a criterion, measuring devices were considered at terminal-buses in which its distance to the nearest junction-bus is longer than 180 meters. It is also important to be highlighted that there is only one section between the junction-bus-67 and the terminal-bus-68. However, the distance between such buses is 1000 meters, therefore, a measuring device was placed at bus-68.

The simulation results obtained for the 134-bus system are analyzed based on the distance error, calculated by (27).

$$\text{Dist. error(m)} = \text{real fault distance-calculated fault distance} \tag{27}$$

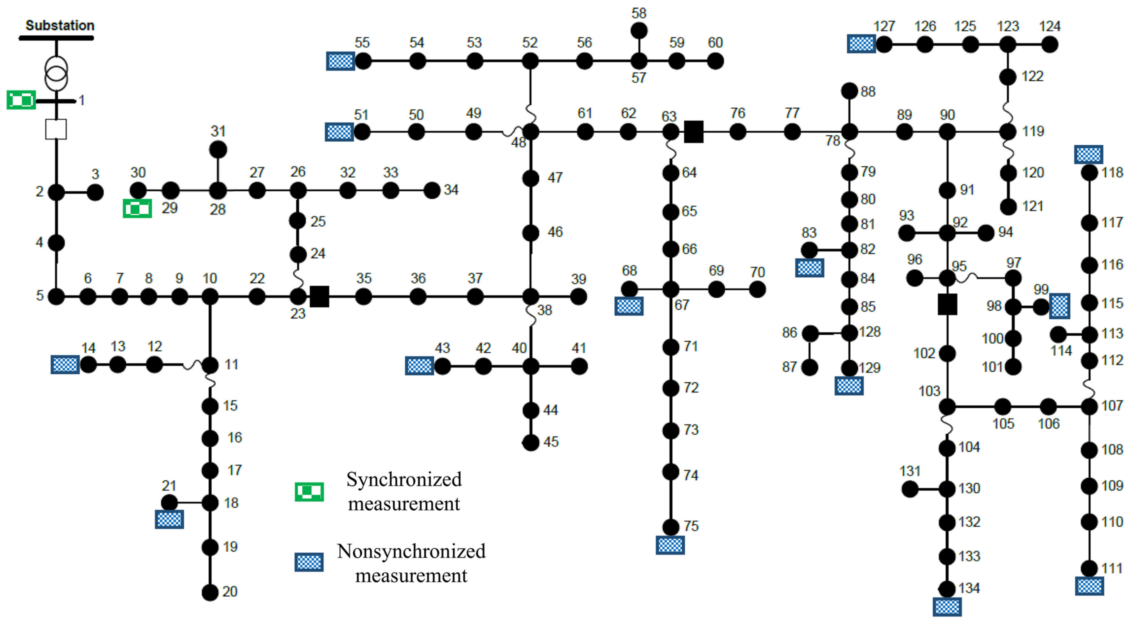

Fig. 15. Diagram of the 134-bus, 13.8 kV distribution system.

TABLE XV
SIMULATION RESULTS FOR THE 134-BUS SYSTEM

DG	Fault type	Fault resistance	Distance Error (m)				
			0-50	51-100	101-150	151-200	201-230
No	LG	1-100 Ω	102	24	6	1	0
Yes	LG	1-100 Ω	72	39	14	7	1

TABLE XVI
DISTRIBUTED GENERATORS INSTALLED IN THE 134-BUS SYSTEM

Bus	DG type	Capacity
3	Asynchronous (ADG)	500 kVA
98	Asynchronous (ADG)	150 kVA
51	Synchronous (SDG)	200 kVA
134	Synchronous (SDG)	100 kVA
58	Photovoltaic (PV)	500 kW
92	Photovoltaic (PV)	250 kW
123	Photovoltaic (PV)	300 kW

The results of the simulation on the 134-bus system are given in Table XV for LG fault type with/without DG presence. For all line sections, an LG fault was simulated at 50%, resulting in 133 fault cases. The DGs data are presented in Table XVI.

From Table XV, very good results were obtained for the 134-bus system with/without DG. In most cases, the faults were located with a distance error from 0 to 50 meters even under DG presence. Only in a case, the method indicated a fault with a distance error between 201 and 230 meters.

The methods proposed in [2], [4], [13], and [14] were also validated on the 134-bus systems, where the errors were also calculated by (27). In [2], in most cases, the distance error also ranges from 0 to 50 meters and, in some cases, an error distance of 220 meters is obtained. For LG faults simulated at 50% of the lines, without considering measurements errors and DG presence, our method presented a mean error of 34.64 meters, while a mean error of 41.2 meters was obtained in [2].

TABLE XVII
SIMULATION RESULTS FOR THE 134-BUS SYSTEM OBTAINED FOR THE PROPOSED METHOD AND REFERENCES [4], [13], AND [14]

Methods	Fault type	Fault resistance	Distance Error (m)				
			0-100	100-200	200-300	300-400	>400
Proposed	LG	1-100 Ω	126	7	0	0	0
[4]		100 Ω	94	33	4	2	0
[13]		100 Ω	105	21	4	2	1
[14]		10 Ω	120	10	2	1	0

On the other hand, the methods proposed in [4], [13], and [14] presented, in most cases, distance errors from 0 to 100 meters, and higher than 400 meters in some cases. LG faults simulation results are presented in Table XVII for different methods without considering measurement errors and DG presence. Therefore, from such comparisons, for LG fault, the method introduced in this paper presents very good performance with/without DG presence.

IV. CONCLUSION

This paper proposes a new method for fault location in distribution networks based on classical circuit analyses. Pre- and during-fault voltages are required at few buses along with the matrix impedance, and load parameters are not required. The fault location process is performed by solving systems of determined equations, which makes the proposed method more technically accessible than the existing methods. The procedure of investigating the subsystems/laterals separately prevents the fault of being located at different laterals. Besides being insensitive to the fault resistance, the method is validated for all fault types. Furthermore, even under DG presence, line parameter errors, voltage measurement errors, and high loading and unbalanced system conditions, the proposed method provides robust location results.

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